ChatGPT

AI Software – Agentic AI Project Ideas for NYC/CUNY Themes

AI for Zero Hunger & Food Equity

NYC and CUNY are already leveraging AI to tackle food insecurity. For example, Brooklyn College will use AI agents to analyze campus pantry usage and optimize food inventory to reduce waste [1]. The NYC Food Policy Center notes over 700 community kitchens and pantries citywide [2], and NYC Open Data provides a "Supply Gap" map of unmet food needs by neighborhood [3]. Recent CUNY research shows younger, lower-income households waste more expensive foods like protein, suggesting targeted AI meal-planning interventions [4].

Project Idea: **AI-Powered Food Redistribution Agent.** An autonomous system that continuously monitors food supply and demand across NYC communities and campus pantries, then plans real-time redistribution. It could scrape inventory and usage data from CUNY food pantry systems and local donation centers, predict shortages or surpluses, and dispatch requests (e.g. via SMS/API) to nearby donors or volunteers. The agent would use an LLM (e.g. GPT-4o or LLaMA) with tool integrations for data retrieval, plus scheduling and routing modules to coordinate pickups and drop-offs. By acting autonomously (adjusting plans as new data arrives), it can reduce waste and ensure perishable items reach hungry communities timely.

- **Data & APIs:** NYC Emergency Food Supply Gap data (NEEDS assessment) [3]; NYC Food Pantry/Community Kitchen locations (FoodHelp.nyc lists ~700 sites) [2]; CUNY campus pantry usage logs (internal or survey data); USDA food bank APIs; local food inventory systems.

- **Tools & Platforms:** LangChain or AutoGPT for agent orchestration; OpenAI GPT-4/GPT-4o or an open LLM with retrieval for planning and communication; Python libraries (Pandas/NumPy) for data analysis; graph or optimization libraries (NetworkX, OR-Tools) for routing; Twilio or Slack API for messaging volunteers.
- **Evaluation Metrics:** Reduction in total food waste (lbs) across participating sites; number of meals redistributed; pantry stock-out events prevented; user satisfaction (donors/recipients); equity measures (are food-short neighborhoods reached more consistently).

AI for Environmental Sustainability

New York City is advancing sustainability through data-driven plans. For example, NYC's Urban Forest Plan aims to expand tree canopy from 23.4% to 30% by 2040 to address heat-equity gaps [5]. The Mamdani administration's 2026 NYCHA Sustainability Agenda sets decarbonization targets (e.g. clean heating for 20,000 apartments, 10,000 induction stoves) [6]. The NYC Climate Dashboard overlays environmental justice communities on city carbon and resilience metrics [7]. These trends point to rich data sources (climate, waste, energy use) that AI can leverage.

Project Idea: Agentic Carbon Footprint and Waste Optimization Tool. A multi-step AI agent that helps campuses and neighborhoods track and reduce carbon and waste. The agent would autonomously fetch data from NYC Open Data (e.g. building energy usage from Local Law 84 benchmarking, DSNY waste collection stats) and campus IoT sensors. It would use these inputs to model current emissions or waste flows (possibly via trained regression or ML models), then iteratively propose interventions (e.g. peak-shaving schedules, increased recycling pickup) and simulate outcomes. For example, using weather and usage predictions, the agent could schedule HVAC pre-cooling times to use cheaper, greener power. It could also analyze recycling contamination rates with computer vision and recommend better bin placements. By chaining these tasks, the agent continuously refines sustainability actions.

- **Data & APIs:** NYC LL84 Energy Benchmarking dataset (building energy/carbon) [8]; DSNY waste/recycling tonnage data; NOAA climate and air quality APIs; NYC Parks tree canopy data; NYC Open Data environmental sensors (e.g. air quality); campus utility meters.
- **Tools & Platforms:** LLMs (GPT-4/GPT-4o or open LLaMA-3) for planning and natural-language reporting; RL frameworks (OpenAI Gym or Ray RLlib) if optimizing control policies; PyTorch/TensorFlow for ML models (e.g. predicting demand); geospatial tools (GeoPandas, QGIS) for mapping tree data; Vision libraries (OpenCV) for waste-sorting prototype.
- **Evaluation Metrics:** Tonnes of CO_2 emissions reduced or energy saved; waste diversion rate increase (% recycled vs landfill); number of intervention suggestions implemented; cost savings on utilities; equity impact (are underserved neighborhoods seeing improvements?).

AI for Justice, Policy & Equity

City policies are moving toward AI oversight and fairness. NYC's GUARD Act (Nov 2025) established an independent "Algorithmic Data Accountability" office, mandated fairness standards and a public registry of city AI tools [9] [10]. At the same time, a Dec 2025 audit found NYC's bias-audit law for hiring tools to be weakly enforced [11]. These developments highlight the need for automated bias detection and transparency tools.

Project Idea: Autonomous Fairness Audit Agent. An AI agent that ingests public-service algorithms or datasets (e.g. city hiring tools, predictive policing scores, or benefits-eligibility models) and autonomously evaluates them for bias. It would use legal fairness criteria (disparate impact, equalized odds) to test model outputs across demographic groups. For example, given a hiring model or synthetic candidates data, the agent could query it repeatedly (using a Python agent that uses a model's API or emulation) and compute bias metrics (e.g. selection rates by race/gender). If bias is found, the agent would suggest mitigating steps (reweighing, data augmentation). The agent could also fetch relevant case law or regulations (NYC Local Law 144 guidelines) and generate a compliance report. This is agentic because it plans multiple analysis steps and adjusts queries based on findings.

- **Data & APIs:** NYC open datasets for equity analysis (311 service requests [12] for language/access complaints; Stop-and-Frisk demographic data; NYC housing discrimination complaints); US Census demographics for reference groups; publicly posted audit reports (LL144 audit summaries); NYC Council AI registry (when available).
- **Tools & Platforms:** Fairness libraries (IBM AIF360, Fairlearn) for bias metrics; LLMs for generating natural-language audit findings; Python for API calls to city datasets or simulation; knowledge-graph tools to connect laws/regulations; a database (SQLite or Neo4j) to store model/test results.
- **Evaluation Metrics:** Reduction in measured bias metrics (disparate impact ratios) before/after; number of ML models audited; adherence to legal standards (LL144 compliance points); stakeholder trust surveys; time saved for compliance teams.

AI for Health & Public Good

NYC's public health data underscores inequities. A Feb 2026 NYC Health Dept study found that Black and Latino New Yorkers have significantly less wealth and correspondingly worse health outcomes than other groups [13], highlighting deep socio-economic health gaps. The city has since launched programs (medical debt relief, new youth clinics) to address these gaps [14]. AI can help bridge these divides by improving access and trust in health services, especially for underserved communities.

Project Idea: Intelligent Health Navigator Agent. A multi-modal AI assistant that helps NYC residents, especially low-income and LEP (Limited English Proficiency) populations, find healthcare and wellness resources. The agent would listen to user queries (voice/text), fetch personal context (age, insurance status), and use NYC Health and Human Services open data (clinic locations, eligibility rules) to recommend appropriate services (free clinics, insurance enrollment, mental health lines). It could proactively analyze public health alerts or city dashboards (e.g. outbreak data, vaccination sites) and reach out via SMS/app push to affected communities. The agent's "agentic" aspect is its continual learning: it tracks user interactions to refine its recommendations, schedules follow-ups, and alerts emergency contacts if health risks are detected.

- **Data & APIs:** NYC Health Department APIs (clinic locations, community health profiles); NY State immunization registry (if accessible); web scrape of HRA benefit eligibility info; CDC/NYC open data on disease outbreaks; CUNY student health center data (if available); language databases for translation.
- **Tools & Platforms:** Transformer models (GPT-4, Google PaLM, or open Mistral) fine-tuned on health Q&A; voice/NLP libraries for multi-lingual support; dialog management (Rasa or LangChain) for conversation flow; secure database for user info; HIPAA-compliant cloud (AWS/GCP) for sensitive data.

- **Evaluation Metrics:** Improvement in appointment/bookings at clinics; survey of user trust/satisfaction; reduction in missed clinic visits; diversity of served populations (language, borough); health outcome proxies (e.g. increase in preventive screenings).

AI for Accessible & Inclusive Education

New York City schools are actively formulating AI policies. In March 2026 the NYC DOE released AI guidelines using a traffic-light system: e.g., AI can be used to generate translations for multilingual learners or adapt materials for students with disabilities—but only under human supervision [15]. At the same time, students and parents debate AI's role in learning [15]. These dynamics create a space for AI tools that make learning more equitable and inclusive.

Project Idea: **Adaptive Learning Tutor Agent.** An AI-powered tutoring platform that personalizes instruction to each student's needs. The agent would converse with the student (voice or text), diagnose their learning style and gaps (e.g. via quiz questions), then generate customized lessons. It would automatically translate explanations into the student's preferred language and adjust reading level or formats (audio, visuals) for accessibility. For example, a math problem description could be narrated and visualized for a student with dyslexia. Crucially, the agent logs its suggestions for teacher review (matching the DOE's "yellow light" supervision requirement [15]). Teachers and tutors can override or refine the agent's guidance. The system continuously learns from student performance, adapting future lessons.

- **Data & APIs:** Open educational resources (Khan Academy APIs, OER Commons); CUNY Blackboard or public course materials; NYC DOE student performance statistics (aggregated); speech-to-text and text-to-speech engines; translation APIs (Google Translate, DeepL).
- **Tools & Platforms:** LLMs (GPT-4o, Llama-3) for content generation and explanation; multimodal models (Meta's LLaVA, Google's Parti) for image-based questions; reinforcement learning to adapt tutoring policy; accessibility libraries (TTS, AAC).

- **Evaluation Metrics:** Student engagement and completion rates; pre- and post-test performance gains; reduction in dropout or tutoring hours; user feedback on comprehension; accessibility compliance (WCAG) and multilingual support coverage.

AI for Housing & Urban Communities

Housing affordability and stability remain urgent issues. A March 2026 NY State Comptroller report highlights that NYC housed 97% of its homeless (≈140,000 people) but unsheltered homelessness still rose 78% since 2019 [16]. The report urges better use of outreach data to target services [17]. Meanwhile, federal policies (e.g. proposed work requirements) threaten NYCHA and Section 8 tenants, underscoring the need for tools that help residents navigate housing resources.

Project Idea: Tenant Support and Housing Access Agent. A proactive AI assistant for renters and the unhoused. For at-risk tenants, the agent would monitor publicly available housing data (NYCHA development status [18], HPD code complaints [19], eviction filings) and alert social workers or tenants when problems arise (e.g. repeated code violations). For people seeking housing, it would intake user profiles (income, family size, needs) and query NYC Housing Connect, NYCHA waiting list updates, and non-profit shelters to match them with available units or support services. It could automate form-filling and legal document retrieval (e.g. Tenant Bill of Rights text) using an LLM. The agent operates with autonomy: if a match or risk is detected, it can initiate outreach (email/SMS) to the user or their caseworker without waiting for manual steps.

- **Data & APIs:** NYC NYCHA Public Housing locations [18]; HPD Housing Maintenance complaints and violations datasets [19]; NYC Housing Connect apartment vacancy API (lottery listings) [20]; 311 housing violation requests; federal and state rent subsidy rules (HUD API if any); legal aid databases (NYCLU data).
- **Tools & Platforms:** Geographic info (Google Maps API, Mapbox) to locate nearest resources; LLM for Q&A and form automation; data analysis (scikit-learn) to predict eviction risk from complaint history; web automation (Selenium or API scripts) to interact with Housing Connect.
- **Evaluation Metrics:** Number of tenants assisted (matched to housing or legal help); decrease in acute housing emergencies; speed of connecting to shelter; improvement in housing stability (fewer evictions); user satisfaction with navigation assistance.

AI for Student Community & Digital Access

NYC is expanding digital inclusion for residents. The **Big Apple Connect** program provides subsidized broadband to NYCHA sites, and its new Digital Learning Hub offers computer skills and AI classes to 333,000 public-housing residents [21]. This focus underscores a citywide push to bring everyone online and technologically literate. CUNY's own digital equity initiatives (e.g. device lending, training) complement these efforts.

Project Idea: **Peer Mentor & Digital Access Agent.** An AI platform to strengthen student community and bridge the digital divide. For onboarding, the agent would automatically connect new CUNY students with peer mentors who share interests or backgrounds (e.g. field of study, first-generation status). It would use chat to introduce mentor-mentee pairs and suggest group study events. Simultaneously, the agent tracks students' access to technology: it could provide automated reminders to apply for Big Apple Connect broadband, locate the nearest campus computer lab, or request device loans. The system acts agentically by autonomously discovering each student's needs (via short surveys or usage patterns) and continuously matching resources without human scheduling.

- **Data & APIs:** CUNY student directory (opt-in data on majors/interests); NYC 311/computer lab locations; device loan inventory APIs; NYC Big Apple Connect signup information; campus event calendars.
- **Tools & Platforms:** Recommendation systems (collaborative filtering) for matching peers; conversational agent framework (e.g. ChatGPT with custom knowledge base) for Q&A on services; scheduling integration (Google Calendar API) for setting up mentor sessions; content delivery network (e.g. mobile app) for notifications.
- **Evaluation Metrics:** Student engagement (attendance at mentor events, logins); reduction in self-reported digital access issues; increase in utilization of campus digital resources; surveys on sense of belonging and connectedness; diversity of participants across programs.

AI for Science, Data & Public Innovation

New York City champions open data and civic technology. In Spring 2026, **Open Data Week** marked 10 years of NYC's Open Data Law, drawing a record 6,700 participants to conferences and hackathons [22]. Agencies offer hundreds of datasets (transport, health, education) through the NYC Open Data Portal, and the recent School of Data convenings highlight a vibrant civic data community [23] [22]. These initiatives create rich material for AI-driven insight tools.

Project Idea: Civic Data Analytics Agent. An intelligent dashboard assistant that turns NYC open data into actionable insights. Users (policymakers or community groups) can query the agent in natural language (e.g. "How has tree canopy coverage changed in Queens vs Bronx over 10 years?"). The agent uses an LLM with connected data tools: it will automatically fetch the relevant datasets from the portal (e.g. PLUTO land

use data, tree inventory [23]), compute the answer (aggregations, charts), and even generate visualizations or slide decks. It can also proactively monitor data streams (like 311 service volumes or Open Data API feeds) and alert communities when anomalies appear (e.g. spike in traffic accidents). By performing these steps (data lookup, analysis, presentation) autonomously, the agent helps turn raw data into community-driven action.

- **Data & APIs:** NYC Open Data APIs (e.g. yellow taxi trips, heatmap geodata); CityStat dashboards; Census/ACS data for demographics; state open data (NY State data.ny.gov) for comparison; libraries like Socrata API.
- **Tools & Platforms:** RAG pipeline (LangChain) combining GPT-4/GPT-4o with Pandas/NumPy for computation; data visualization (Plotly/D3) for charts; FastAPI or Streamlit for interactive UI; vector databases (Weaviate) for storing scraped data context.
- **Evaluation Metrics:** Accuracy of answers on held-out data queries; number of insights generated; user satisfaction in surveys; actual policy or community decisions influenced by the tool; performance metrics (response latency, API costs).

Agentic AI Toolchain (Across Projects): In all projects, we can leverage modern agent frameworks. For planning and reasoning, large language models (OpenAI GPT-4/GPT-4o, Google Gemini, or open models like LLaMA-3/Vicuna) provide the “brains”. We’ll use chain-of-thought chaining (via LangChain or AutoGPT) to decompose tasks. For example, AutoGPT-like agents can iteratively browse data and update analysis. Reinforcement learning libraries (OpenAI Gym, RLlib) can train control policies (e.g. for efficient routing in redistribution). Standard AI/ML libraries (TensorFlow, PyTorch) support specialized models (vision, speech). We’ll deploy on cloud platforms (AWS/GCP) or campus servers with appropriate APIs.

Each solution will prioritize existing NYC and CUNY data sources, public APIs, and open frameworks. We will measure success using task-specific metrics (waste reduced, bias mitigated, learning improved, etc.) and community feedback. This ensures our agentic AI prototypes are innovative **and** grounded in real NYC/CUNY needs.

Sources: Recent CUNY and NYC press releases, policy announcements, and open-data portals [1] [3] [2] [4] [5] [6] [7] [24] [9] [13] [15] [16] [18] [21] [23] [22] provide the factual basis for these ideas. Each project concept builds on these trends and resources.

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